Srikanth Gedela

SCIENTIFIC INSTRUMENTATION | STABLE ISOTOPE MASS SPECTROMETRY | INSTRUMENT DEVELOPMENT

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Professional Profile

Scientific instrumentation specialist with experience across stable isotope ratio mass spectrometry, analytical laboratory systems, electronics R&D and marine instrumentation. Combines practical IRMS experience with an electronics and embedded-systems foundation. Skilled in diagnosing complex instrument faults, restoring and optimizing analytical systems, supporting research workflows, and adapting hardware and automation to meet laboratory needs. Particularly interested in custom inlet development, vacuum and gas-handling systems, and research instrumentation.

Technical Expertise

Stable isotope and mass spectrometry: Stable isotope IRMS; Isoprime 100; Isoprime precisION; GasBench; elemental analyzers; IRMS inlets; GC–IRMS workflows; carbon and nitrogen isotope analysis.
Instrument troubleshooting and repair: Board-level electronic fault diagnosis; schematic-based troubleshooting; component replacement; instrument restoration; preventive maintenance; calibration; vacuum systems; gas handling.
Custom instrumentation and automation: Custom inlet process automation and development; sample handling; solenoid-valve control; instrument upgrades; prototype development; laboratory process automation.
Electronics and embedded systems: Analog and digital electronics; PCB design and testing; C and microcontroller programming (PIC/8051); sensors; signal conditioning; instrumentation testing and debugging.
Marine and environmental instrumentation: CTD profilers; water samplers; biogeochemical sensors; FRRF; pCO_2 systems; aerosol sampling; instrument deployment and recovery; oceanographic field operations.
Engineering tools: ORCAD; AutoCAD; Alibre; MATLAB; MPLAB; MicroC; oscilloscopes; logic analyzers; signal generators.

Professional Experience

Isotope Mass Spectrometer Specialist | September 2025–Present

Department of Earth, Environmental and Planetary Sciences (DEEPS), Brown University, Providence, Rhode Island, USA

- Support stable isotope mass spectrometry and associated laboratory instrumentation in a university research environment, including GasBench coupled to an Isoprime 100.
- Perform board-level troubleshooting on a Thermo Scientific NEOMA instrument, using available electrical schematics to trace faults, identify defective components and repair the affected board, restoring it to full operational function.
- Troubleshoot a Coherent CO_2 laser system experiencing reduced output; sourced a compatible replacement laser unit, integrated it into the existing system and independently learned the optical configuration and beam path.
- Aligned the invisible infrared laser beam through the optical system and iteratively refined the alignment to recover and improve laser energy output.
- Contribute to instrument maintenance, operational reliability and analytical system improvement, applying an instrumentation-development perspective to laboratory needs.

Assistant Service Manager | January 2019–2025

Elementar India Pvt. Ltd., India

- Led and supported a team of approximately eight service engineers, coordinating technical troubleshooting and customer support for analytical instrumentation.
- Provided operation support, user training, maintenance and fault diagnosis for stable isotope ratio mass spectrometers, including Isoprime 100 and Isoprime precisION, elemental analyzers and associated IRMS inlets.
- Supported GC–IRMS and other analytical systems, including gas chromatography and related laboratory equipment.
- Coordinated service activities and customer cases, guided colleagues through complex troubleshooting, and supported service quality across customer laboratories.

Project Assistant / Instrumentation Engineer | April 2012–2018

CSIR–National Institute of Oceanography (NIO), Visakhapatnam, India

- Supported stable isotope analysis of carbon and nitrogen in solid, liquid and gaseous samples using IRMS coupled with elemental analysis and GasBench systems.
- Operated, maintained, troubleshot and calibrated analytical instruments, including IRMS, GC–IRMS, gas chromatography, HPLC and associated laboratory systems.
- Supported marine and environmental instrumentation, including CTD profilers, water samplers, biogeochemical sensors, FRRF and pCO_2 systems.
- Participated in oceanographic expeditions and coastal field operations, supporting instrument deployment, recovery, data acquisition and reporting to research teams.

- Developed and adapted laboratory instrumentation to meet research requirements, including coulometer acidification-unit automation using solenoid pinch-off valves, an absolute-wind aerosol sampler, and sensor-integration hardware.
- Provided technical guidance to research staff, scholars and visiting users, and supported laboratory procedures and instrument readiness.

R&D Engineer (Electronics) | January 2010–March 2012

Zen Technologies Ltd., Hyderabad, India

- Worked on the design, development, testing and maintenance of electronic and embedded systems for simulation and training products.
- Developed and tested firmware using PIC and 8051 microcontrollers.
- Designed and debugged electronic circuits and PCBs using ORCAD and laboratory test equipment, including oscilloscopes, logic analyzers and signal generators.
- Tested sensors, power supplies, connectors and signal-conditioning circuits; prepared bills of materials and technical documentation.
- Contributed across project stages from requirements and prototype development through validation and maintenance.

Production Engineer (Instrumentation) | May 2008–January 2010

Nucleonix Systems Pvt. Ltd., Hyderabad, India

- Coordinated production activities for nuclear instrumentation products, including production scheduling and material and equipment planning.
- Performed calibration and testing of completed instruments and supported installation, demonstration and customer service.
- Supported product promotion at technical workshops and gathered customer requirements for product upgrades and new development.

Design Assistant (Electronics), part-time | April 2004–January 2008

Techcel Electronics and Services, Visakhapatnam, India

- Assembled, tested and troubleshot custom electronic PCBs and prototype systems.
- Worked with analog and digital circuits, schematics and technical drawings, using oscilloscopes, signal generators and power supplies.

Selected Instrumentation & Development Work

- **Automated Acidification & Purging System (AAPS) for UIC Coulometer — Design and Development**

  Designed and developed an automated system to control sample and acid intake, followed by purging for preset durations during coulometric analysis.

  Integrated three-way solenoid pinch valves for non-invasive flow control, flow measurement, programmable digital timing and pre-/post-scrubbers.

  Developed the system to automate the acidification and purging workflow and reduce manual intervention during analysis.

- **Wind-Based Absolute Reference Aerosol Sampler (ARAS)**

  Designed and developed a wind-based aerosol collection system for sampling aerosols under different wind-direction conditions.

  Developed a multi-channel sampling arrangement with dedicated collection chambers for four wind directions (NE, ES, SW and WN), along with an all-direction sampling channel.

  Integrated a vacuum pump and solenoid-valve arrangement to control airflow through the selected sampling channels.

  Incorporated a wind vane and an electronic control unit with drive, GSM communication, power supply and data-logging functions.

- **Custom inlet process automation and development —** Exploration and development of laboratory inlet concepts, sample handling and automation for isotope-ratio analytical workflows.
- **Laser-assisted elemental analysis concept —** Developing a research concept for laser-based sample conversion and gas analysis, integrating controlled gas handling, drying, gas chromatography and thermal-conductivity detection. This is an ongoing concept-development activity.
- **Laboratory and marine instrument development —** Examples include coulometer acidification automation, an aerosol sampler, an electronic sample pipettor and sensor-integration structures for marine deployments.
- **Embedded electronics prototypes —** Earlier microcontroller projects included monitoring and display systems, RF-based detection, ECG instrumentation and autonomous robotics.

Education

M.Sc. in Instrumentation, Andhra University, India — 2008.

B.Sc. in Electronics, Andhra University, India — 2006.

Training & Professional Development

- Intensive training in in-situ nutrient monitoring, Green Eyes LLC, Easton, Maryland, USA (15 November–13 December 2017).
- “Instrumenting our Oceans for Better Observation: A Training Course on Biogeochemical Sensors,” Sven Lovén Centre for Marine Sciences, Kristineberg, Sweden (22 June–1 July 2015).
- Training on sensors and ocean data collection, National Institute of Ocean Technology (NIOT), India (9–11 February 2016).
- Training at Bhabha Atomic Research Centre (BARC) on USB-based multi-channel analyzer data acquisition and liquid scintillation counting systems; participated in technology-transfer activities.

Selected Publication

Vipin, P., Kankan Sarkar, S. G. Aparna, D. Shankar, V. V. S. S. Sarma, D. G. Gracias, M. S. Krishn, G. Srikanth, R. Mandal, E. P. Rama Rao, and N. Srinivasa Rao (2015). “Evolution and sub-surface characteristics of a sea-surface temperature filament and front in the northeastern Arabian Sea during November–December 2012.” Journal of Marine Systems. https://doi.org/10.1016/j.jmarsys.2015.05.003

Additional Information

Languages: English, Hindi and Telugu.